

The asymptotic complexity of the mirror reconciliation protocol

Canonical source; supersedes the retired markdown original ([results/mirror-complexity.md](#)).

A first-principles derivation of the per-side computation, bandwidth, and latency of **rumors'** mirror protocol, written for a reader who knows basic probability and big-O but has never seen this codebase. Every modeling assumption is pinned to a code location; every approximation is named where it is used. All file references are relative to the repository root (this worktree); line numbers are as of commit **a8a3b7a**.

Headline results (derived in §6, constants from the code):

Axis	Result (expected, uniform leaves)	Empirical fit it refines
Per-side computation	$\Theta(\min(256 \cdot D, N) \cdot (1 + \log_{256}(N/D)))$ child-hash comparisons; equivalently $\Theta(D \cdot (1 + \log(N/D)))$ disputed nodes visited, $\times 256$ sibling fan-out each	$\sqrt{C \cdot D}$: a true upper bound (§7), loose by up to $\sqrt{C/D}/\ln(C/D)$
Bandwidth	$\Theta(D)$ message bodies + the same $\min(256 \cdot D, N) \cdot (1 + \log_{256}(N/D))$ hash entries (~ 36 – 64 B each) + ≤ 36 frames of framing	$\sim D$: correct up to the slowly-varying log factor
Latency	$\frac{1}{2} \cdot \log_{256}(2 \cdot D \cdot N) + O(1)$ round trips, hard ceiling ≈ 18 RTT (36 frames), floor 1 RTT after the preamble for identical versions	$\log_{256}(C + D)$: same at $D \approx N$, $\sim 2\times$ high for $D \ll N$ (both tiny)

where $N = C + D$. One meta-finding up front, developed in §7: at every realistic scale ($N \leq 10^7$), the *exact* level-by-level sum tracks $\sqrt{C \cdot D}$ to within a factor of ~ 2 – 4 across almost the whole (C, D) plane. The smooth $D \cdot \log(N/D)$ form is the asymptote; the discrete base-256 staircase underneath it locally mimics a square root. The empirical fit was not an accident, and wall-clock benchmarks at these scales genuinely cannot refute it — §9 gives measurements that can.

Source-of-data flag. `benches/hash_recon.rs` is *not* a reconciliation benchmark: it measures worst-case root-hash *reconstruction* (32 iterated 8 KiB BLAKE3 wraps vs. 34-byte commitments), motivating the per-node hash memo. The reconciliation measurements live in `benches/gossip_grid.rs` and `benches/in_memory.rs` over the divergence grid in `benches/support/grid.rs`. Everything below assumes the $\sqrt{C \cdot D}$ fit came from that grid.

1 The system, in one page

Each replica holds a **sparse Merkle radix trie**: branching factor 256, fixed depth 32, one path byte per level (`src/tree.rs:18-26`, `src/tree/typed/height.rs:110-117`). A leaf’s 32-byte path is `blake3(blake3(version) || blake3(value))` — the *version is folded into the path* (`src/tree/typed/path.rs:32-44`), so every insert, even of identical content, lands at a fresh, computationally-uniform position. Single-child runs are path-compressed into a prefix byte string on the node (`src/tree/typed/untyped.rs:25-47`, invariant “branches have ≥ 2 children” at `untyped.rs:78-81`, 128–150). Each node lazily memoizes its subtree hash (`untyped.rs:266-288`) and its version ceiling/floor (`untyped.rs:298-360`); the memos survive copy-on-write clones (`untyped.rs:100-123`). Hashing is compression-invariant: a compressed prefix byte wraps the child hash exactly as a materialized one-child branch would (`src/tree/typed/hash.rs:54-79`, `untyped.rs:268-287`). Leaves hash to a constant (`hash.rs:45-51`) — all leaf-distinguishing content lives in the *path*.

The **mirror protocol** reconciles two replicas’ trees over a wire. After a 25-byte raw preamble (`src/tree/traverse/mirror/remote.rs:11-17`) and a framed version handshake, equal versions terminate immediately (`src/tree/traverse/mirror/local.rs:230-241`, 268–281; `remote.rs:279-285`). Otherwise both sides run a synchronized descent. Each side keeps a *zipper* of per-height levels (`src/tree/typed/levels.rs`); each message a side sends descends its own zipper by **two heights** (`local.rs:395-415`; `partition.rs:354-356`), and because the two sides alternate strictly, the conversation’s frontier advances **one depth per message, two per round trip**.

Each steady-state message carries three channels, per the wire format in `src/tree/traverse/mirror/message.rs:50-70` and the asymmetry matrix at `local.rs:46-58`:

Channel	Content	Receiver’s action
uncertain	(<i>prefix, hash</i>) of the sender’s disputed-frontier subtrees	merge-join against own children: equal hash $\Rightarrow$ drop; differ $\Rightarrow$ explode grandchildren (recurse); we-lack $\Rightarrow$ request; they-lack $\Rightarrow$ provide
requested	prefixes the sender lacks entirely	answer next message by exploding that node’s children into providing (<code>partition.rs:91-142</code>)
providing	whole subtrees, full structure on the wire (<code>message.rs:50-58</code>)	insert at the named prefix (<code>partition.rs:57-85</code>)

All per-round set operations are single linear merges over sorted flat vectors (`partition.rs:100-115`, 199–203; `src/tree/typed/levels/level.rs`), so per-round work is linear in the frontier and message sizes — no hidden log factors beyond the `imbl OrdMap` child maps (an $O(\log 256)$ constant per child operation, absorbed below).

Termination is in-band: a side is done when its outgoing **requested** and **uncertain** are both empty (`partition.rs:393-410`; `remote.rs:34-43`, 364–374, 398–409); a sender that just emitted an empty-empty message does not even await a reply (`remote.rs:398-409`). The schedule has a hard ceiling — `mirror_connected` (`src/tree/traverse/mirror.rs:126-137`) is `initiator + responder + open_initiator + a seq! loop of 14 exchange pairs +`

`close_initiator` + `complete_responder` + `complete_initiator`. Counting frames: 2 handshake + `Initiate` (depth 0) + `Opening` (depth 1) + `Exchange` (depth 2) + 28 exchanges (depths 3–30) + one more responder exchange (depth 31) + `Closing` + `Complete` = **36 frames maximum**, covering all 32 byte-levels (the chain lengths are pinned at the type level in `src/tree/traverse/mirror/protocol.rs:477–488`).

Deletions leave no tombstones. Every subtree shipped through `providing` is first filtered by `Unknown::unknown` (`src/tree/traverse/unknown.rs`): a subtree whose version *ceiling* is $\leq$ the counterparty’s version vector is something they have already seen — its absence on their side means they deleted it — so it is dropped in $O(1)$ (one memoized-ceiling comparison, `unknown.rs:28–34`), *and the local copy is dropped too* (the Left arm keeps a node only if it survives the filter, `partition.rs:232–239`). A floor check keeps wholly-unknown subtrees intact without rebuilding them (`unknown.rs:38–52`).

2 Model and assumptions

- **(A1) Uniform leaf placement.** Leaf paths are BLAKE3 outputs over *(version, value)* (`path.rs:32–44`), and versions are unique per action (`src/tree.rs:299–318`). In the random-oracle model the N leaf paths are i.i.d. uniform 32-byte strings, *regardless of insertion order, batching, or payload skew* — an adversary would need structured BLAKE3 preimages to cluster them. This is the load-bearing probabilistic assumption; see §10.4.
- **(A2) Parameters.** Branching $b = 256$, depth 32 (`tree.rs:21`, `height.rs:110–117`). Hash size $h = 32$ bytes (`hash.rs:17`).
- **(A3) Warm memos.** Subtree hashes and version bounds are memoized reads (`untyped.rs:266–288`); we count a hash *comparison* as $O(1)$. First-touch costs after local mutation are a separate, additive term (§10.1) — the benches warm them in untimed setup (`benches/gossip_grid.rs:119–160`, `src/tree.rs:182–196`).
- **(A4) Dispute drives descent.** Only the *Both-and-hashes-differ* cell of the asymmetry matrix recurses (`partition.rs:243–254`). The Left cell ships a whole subtree in the same message; the Right cell is requested and answered whole one message later (`partition.rs:91–142`). So one-sided differences cost $O(1)$ additional messages once they separate from two-sided content.
- **(A5) Two heights per send, one depth per message.** Verified at `local.rs:395–415` and the alternation in `mirror.rs:126–137`.
- **(A6) No level skipping.** Path compression does **not** shorten the descent: exploding a compressed node pops exactly one prefix byte and yields a singleton child map (`untyped.rs:156–167`). Compression saves hashing and memory, never rounds.
- **(A7) No hash collisions.** A false uncertain match (probability $\leq \text{pairs} \cdot 2^{-256}$) would silently prune; ignored.
- **(A8) Wire sizes.** An uncertain entry at depth k is k raw prefix bytes + 32 hash bytes (`prefix.rs:127–156`, `message.rs:61–63`); a `providing` node is its full borsh structure: per leaf, a version, the payload, and ≤ 32 compressed-prefix bytes (`message.rs:21–58`, `untyped.rs:401–438`). Frames carry a 4-byte length prefix (`remote.rs:26–33`).

3 Notation

Symbol	Meaning
b	branching factor, 256
C	live messages present on both sides
D	messages present on exactly one side, summed over both sides ($D = D_L + D_R$); redacted-on-one-side leaves count (§10.3)
N	$C + D$, the size of the union tree
k	depth = prefix length in bytes, $0 \leq k \leq 32$; height $H = 32 - k$
Δ_k	number of disputed depth- k prefixes: prefixes containing ≥ 1 differing leaf <i>and</i> ≥ 1 leaf on the counterparty's side
f_k	expected occupied-children count of a depth- k node, $\approx \min(b, \max(1, N/b^k))$
j_D, j_N	$\log_b D, \log_b N$ (depths where b^k crosses D, N)
k^*	separation depth: deepest disputed prefix
W	per-side work / total uncertain entries: $\sum_k \Delta_k \cdot f_k$
$\bar{s}, s_v$	mean payload size; serialized Version size (not $O(1)$, §10.5)

Logs are base $b = 256$ unless written ln. One decimal decade is $\log_{256} 10 \approx 0.415$ levels; one level spans 2.4 decades. Keep that coarse quantization in mind — it is the source of the staircase in §7.

4 Warm-up: $D = 1, N = 10^6$

One side holds a single extra leaf; the trees otherwise agree on 10^6 messages. Trace the protocol:

1. **Handshake** (2 frames): versions differ, descend.
2. **Initiate** (depth 0): one root hash. Differs.
3. **Opening** (depth 1): the responder *unconditionally* explodes its root and lists all its children (local.rs:333–353) — at $N = 10^6$, all 256 slots are occupied ($\approx 3,906$ leaves each): **256 entries**.
4. The initiator merge-joins: 255 child hashes match and drop; the one slot containing the extra leaf differs (Both-case), so its 256 grandchildren are exploded and listed: **256 entries** at depth 2.
5. The responder finds one depth-2 hash differing; that node has $\approx 10^6/256^2 \approx 15$ occupied children: $\approx$ **15 entries** at depth 3.
6. At depth 3 the extra leaf sits alone in its slot with probability $\approx 1 - N/256^3 \approx 0.94$: a **Left case**. The whole leaf (a path-compressed spine node carrying its remaining ≈ 29 path bytes, version, and payload) ships in **providing**; nothing is exploded; the outgoing **uncertain** and **requested** are empty $\Rightarrow$ **Done**, in-band.

Totals: $\approx 256 + 256 + 15 \approx 527$ hash entries on the wire and compared (~ 18 KB), 7–8 frames ≈ 4 round trips including the preamble, one message body. Three things to notice, which the general analysis formalizes:

- The *visited disputed nodes* number only ≈ 3 (one per level until separation) — $D \cdot (1 + \log_b(N/D))$ with $D = 1$.
- But each visited node costs its **full sibling fan** (≤ 256 entries): the per-node constant is $f_k \approx \min(b, N/b^k)$. This factor-256 is the price of the radix-256 tradeoff: few rounds, fat rounds.
- Separation happened at depth $\approx \log_b(D \cdot N) = \log_b 10^6 \approx 2.5$ — not $\log_b N$ levels of *messages saved*, because termination is the in-band emptiness predicate, not bottoming out at depth 32.

5 Two lemmas

Lemma 1 (prefix occupancy / sharing). *Throw m uniform leaves into the b^k bins at depth k . The expected number of occupied bins is $b^k \cdot (1 - (1 - b^{-k})^m)$, which is $\Theta(\min(b^k, m))$ — at least $(1 - 1/e) \cdot \min(b^k, m)$, at most $\min(b^k, m)$.*

Proof sketch. Linearity of expectation over bins; $(1 - 1/x)^x \leq 1/e$. Applied to the D differing leaves, this bounds how many distinct ancestors they have at each depth: their root-to-leaf paths share prefixes exactly while $b^k < D$, and are essentially disjoint below. Summing, the union of the D differing root-to-separation paths has $\sum_k \min(b^k, D) = \Theta(D \cdot (1 + \log_b(N/D)))$ distinct nodes (the geometric head contributes $\Theta(D)$; the plateau contributes D per level from j_D to the dispute cutoff of Lemma 2). $\square$

Lemma 2 (dispute persistence). *A depth- k prefix is disputed — i.e. recursed into rather than resolved — iff it contains a differing leaf and the counterparty owns at least one leaf under it (A4: otherwise it is a Left/Right case and resolves in $O(1)$ messages). Under A1,*

$$\mathbb{E}[\Delta_k] \leq \min(b^k, D, 2 \cdot D \cdot N \cdot b^{-k})$$

and $\Pr(\text{any dispute survives depth } k) \leq 2 \cdot D \cdot N \cdot b^{-k}$.

Proof sketch. The first two bounds are Lemma 1 applied to the differing leaves. For the third: a dispute at depth k requires some differing leaf d and some leaf x on the side opposite d 's holder to share a k -byte prefix. Each such (ordered) pair shares it with probability b^{-k} (uniform, independent paths), and there are at most $D_L \cdot (C + D_R) + D_R \cdot (C + D_L) \leq 2 \cdot D \cdot N$ pairs. Union bound / linearity. $\square$

Corollary (separation depth). $k^* \leq \log_b(2 \cdot D \cdot N) + t$ with probability $\geq 1 - b^{-t}$; $\mathbb{E}[k^*] \leq \log_b(2DN) + \frac{b}{b-1} \cdot \frac{1}{\ln b} \approx \log_b(2DN) + 1$. *The fixed 32-level ceiling is unreachable probabilistically: hitting it would need $D \cdot N \approx 256^{30}$. (It exists because paths are 32 bytes; even a full-depth descent is correct — equal full paths imply equal (version, value) by construction, **tree.rs:22–26**.)*

Lemma 3 (fan-out). *A depth- k node in the union tree of N uniform leaves has expected occupied-children count $f_k = \Theta(\min(b, \max(1, N/b^k)))$, and the total children over any set of Δ depth- k nodes is $\leq \Delta \cdot b$ always and $\Theta(\Delta \cdot f_k)$ in expectation (children counts across disjoint subtrees are negatively associated; we only ever need linearity). Caution (Jensen): $\mathbb{E}[\min(X, Y)] \leq \min(\mathbb{E}X, \mathbb{E}Y)$, so products of the min envelopes used below are upper envelopes of expectations; the matching lower bounds hold within the $(1 - 1/e)$ constants of Lemma 1.*

6 The level-by-level sum

Per round at depth k , the side processing the counterparty's **uncertain** does (all single-pass, sorted merges — `partition.rs:159-304`):

- drain its frontier once: $O(|\text{frontier}_k|)$;
- for each disputed parent, merge-join its own occupied children against the listed hashes: $O(\text{own fan} + \text{listed entries})$, each comparison a memoized-hash `memcmp` (A3);
- explode each hash-differing child's grandchildren into the next level and emit them as the next **uncertain** (`partition.rs:243-254, 364-373`).

So the entries *listed at depth k* are exactly the occupied children of the disputed depth- $(k-1)$ parents, and each side's per-session work, the total **uncertain** traffic, and the zipper/collapse churn (`levels.rs:148-197`, linear in total level entries) are all Θ of the same master sum:

$$W = \sum_{k=0}^{32} \Delta_k \cdot f_k \approx \sum_k \min(b^k, D, 2DN \cdot b^{-k}) \cdot \min(b, \max(1, N/b^k))$$

Evaluate by regimes (write $j_D = \log_b D$, $j_N = \log_b N$; assume first $D \leq N/b^2$ so all three regimes are non-empty):

Regime	Depths	Term	Contribution
Head (paths shared)	$k \leq j_D$	$b^k \cdot b$	geometric, $\approx b \cdot D \cdot (1 + 1/b + \dots) \approx b \cdot D$
Plateau (paths distinct, fans saturated)	$j_D < k \leq j_N - 1$	$D \cdot b$	$b \cdot D \cdot (\log_b(N/D) - 1)$
Tail (fans thin out, disputes die)	$k > j_N - 1$	$D \cdot N \cdot b^{-k} \cdot O(1)$	geometric, $\approx \Theta(D \cdot b)$ at the shoulder, then $\div 256$ per level

Theorem 1 (per-side computation). *Under A1–A6, the expected per-side work (hash comparisons + level-entry handling, each $O(1)$) and the expected total **uncertain** entry count are*

$$W = \Theta(\min(b \cdot D, N) \cdot (1 + \log_b(N/D)))$$

- for $D \leq N/b$: $W = \Theta(b \cdot D \cdot (1 + \log_b(N/D)))$ — the prompt-level form $\Theta(D \cdot (1 + \log(N/D)))$ counts disputed nodes visited; the protocol pays a $\times b$ sibling fan on each (Lemma 3, §4);
- for $N/b < D \leq N$ (heavily diverged / disjoint): the plateau vanishes and the peak term is $\Theta(N)$; $W = \Theta(N) = \Theta(D) \cdot O(b)$ — linear;
- additive $\Theta(D \cdot (1 + s_v))$ for absorbing/filtering the provided subtrees and the final collapse, and $\Theta(1)$ per round of fixed pipeline cost. $\square$

Cross-checks: for $D = 1$, $N = 10^6$ the log factor sharpens from the envelope's $\log_b(N/D)$ to $\log_b(N/b)$, because the per-level fan f_k saturates at b only down to depth $j_N - 1$ and decays geometrically below — so the sum is b per level over the $j_N = 1 + \log_b(N/b)$ saturated levels, $W \approx 256 \cdot (1 + \log_b(N/b)) \approx 640$, vs. the §4 hand count of 527 ✓ (the residue is the envelope billing the thinned final fraction of a level at the full b where the tree delivers ≈ 15). Billing all $1 + \log_b(N/D) \approx 3.5$ levels of descent at full fan, as the Θ -form read literally would, gives ≈ 894 — still a correct upper envelope, but the descent below depth $j_N - 1$ is exactly where the fans thin. Disjoint trees ($C = 0$, $N = D$) give $\Theta(N)$ ✓ (matches the one-shot character of bootstrap, §10.2).

Theorem 2 (bandwidth). *Total bytes on the wire, both directions (each depth's **uncertain** is listed by exactly one side — the alternation splits levels between them):*

$$\begin{aligned}
\text{bytes} &= W \cdot (h + \bar{k}) && [\text{uncertain: 32-byte hash} + k\text{-byte prefix, } \bar{k} \leq j_N + 1] \\
&+ \Theta(D) \cdot (\bar{s} + s_v + \leq 32) && [\text{providing: payloads, versions, compressed spines}] \\
&+ O(D) \cdot \bar{k} && [\text{requested: one prefix per separated they-only subtree}] \\
&+ \leq 36 \cdot 4 + 2 \cdot 25 + 2 \cdot s_v && [\text{framing, preamble, handshake}]
\end{aligned}$$

In Θ -terms: $\Theta(D \cdot (\bar{s} + s_v)) + \Theta(\min(b \cdot D, N) \cdot (1 + \log_b(N/D)) \cdot h)$. The empirical “ $\sim D$ ” is right in its D -scaling; the per- D constant is the interesting part: at $N/D = 10^2 \dots 10^4$ the smooth envelope prices the hash channel at ~ 10 – 40 KB per differing message (e.g. §4: 18 KB to move one message). That envelope is tight only for $D \lesssim b$; §8 prices the channel exactly and shows the staircase discounts it to ~ 0.3 – 6 KB once $D \gg b$. For small payloads, **hash traffic dominates bandwidth**; bodies dominate only when $\bar{s} \gtrsim 36 \cdot W/D$ bytes (the crossover is solved in §8). □

Theorem 3 (latency). *Messages strictly alternate and the conversation advances one depth per message (A5). Disputes empty at depth k^* (Lemma 2 corollary), plus $O(1)$ messages to flush the final **requested** $\rightarrow$ **providing** exchange; the in-band predicate then stops the session (A6 — no draining of the remaining schedule):*

$$\text{RTT} = \frac{1}{2} \cdot k^* + O(1) = \frac{1}{2} \cdot \log_{256}(2 \cdot D \cdot N) + c, \quad c \approx 3\text{--}4$$

(preamble ≈ 1 , handshake 1, **Initiate/Opening** 1, closing tail ~ 1), hard-ceilinged at 36 frames ≈ 18 RTT, floored at ~ 2 RTT total for equal versions. Path compression does not reduce this (A6). Numerically the descent portion is tiny at any scale — 1.8 levels ≈ 1 RTT for $(N, D) = (10^6, 1)$ up to 6.1 levels ≈ 3 RTT for $(10^7, 10^7)$; the $O(1)$ dominates. The empirical $\log_{256}(C + D)$ agrees at $D \approx N$ (where $\frac{1}{2} \log DN = \log N$) and is $\approx 2\times$ the true descent depth for $D \ll N$, a difference of < 2 RTT at any realistic scale — unfalsifiable from timing, trivially checkable by counting frames (§9). □

7 Against the $\sqrt{C \cdot D}$ fit

Proposition 1 (the fit is a true upper bound). *For $1 \leq D \leq C$:*

$$D \cdot \ln(C/D) \leq \frac{2}{e} \cdot \sqrt{C \cdot D}, \quad \text{with equality iff } D = C/e^2.$$

Proof. Set $x = D/C \in (0, 1]$ and $g(x) = \sqrt{x} \cdot \ln(1/x)$. On $(0, 1]$, $g'(x) = (\ln(1/x) - 2)/(2\sqrt{x})$ vanishes only at $x = e^{-2}$, a maximum, with $g(e^{-2}) = 2/e$. Since $\sqrt{CD} = C\sqrt{x}$,

$$D \cdot \ln(C/D) = C \cdot x \cdot \ln(1/x) = (C\sqrt{x}) \cdot (\sqrt{x} \cdot \ln(1/x)) = \sqrt{CD} \cdot g(x) \leq \frac{2}{e} \cdot \sqrt{CD},$$

with equality iff $x = e^{-2}$, i.e. $D = C/e^2$. $\square$

In base 256: $D \cdot \log_{256}(C/D) \leq (2/(e \cdot \ln 256)) \cdot \sqrt{CD} \approx 0.133 \cdot \sqrt{CD}$. With Theorem 1's constants ($b/\ln b \approx 46.2$), for $D \leq C$:

$$W \leq b \cdot D \cdot (2 + \ln 2) + \frac{2}{e} \cdot \frac{b}{\ln b} \cdot \sqrt{CD} \approx 690 \cdot D + 34 \cdot \sqrt{C \cdot D} = O(\sqrt{C \cdot D}),$$

since $D \leq \sqrt{CD}$ when $D \leq C$. So the empirical $\sqrt{C \cdot D}$ is a *correct* upper bound on per-side computation — but loose: the ratio $\sqrt{CD}/(D \cdot \log(C/D)) = \sqrt{C/D}/\log(C/D)$ grows without bound as $D/C \rightarrow 0$, and the bound is tight (constant-factor) only in the band $D/C \in [\sim 10^{-2}, \sim \frac{1}{2}]$ where $\sqrt{x} \cdot \ln(1/x)$ stays within $1.5\times$ of its max.

Why the fit nevertheless matched the data: the staircase. The smooth form $b \cdot D \cdot (1 + \log_b(N/D))$ is the envelope of a *staircase*: each level of the sum spans a factor of $256 \approx 2.4$ decades. Inside one span the dominant term is $t_k = D \cdot N/b^k$ — **linear in N at fixed D** — and the term above it, $\min(b, D) \cdot b^k$ -shaped, is *flat* in D once $D > b$. The exact sum's local log-log slopes therefore wobble around the sqrt's 0.5 instead of sitting at the asymptotic 0 (in N) and 1 (in D):

Exact-sum values of W (children examined $\approx$ **uncertain** entries; computed from the master sum, expectations, constants $\pm 2\times$ from occupancy corrections), against $\sqrt{C \cdot D}$:

N	D	W	$\sqrt{C \cdot D}$	$W/\sqrt{CD}$	k^*	RTT (descent + c)
10^4	1	295	100	3.0	1.8	~ 3.9
10^4	10^2	4.2 k	1.0 k	4.2	2.6	~ 4.3
10^4	10^4	26 k	10 k	2.6	3.6	~ 4.8
10^5	10^2	26 k	3.2 k	8.2	3.0	~ 4.5
10^5	10^3	67 k	10 k	6.8	3.5	~ 4.7
10^6	1	0.53 k	1.0 k	0.53	2.5	~ 4.2
10^6	10^2	27 k	10 k	2.7	3.5	~ 4.7
10^6	10^3	81 k	32 k	2.6	3.9	~ 4.9
10^6	10^4	220 k	100 k	2.2	4.3	~ 5.1
10^6	10^5	1.08 M	316 k	3.6	4.7	~ 5.3
10^7	1	0.67 k	3.2 k	0.21	2.9	~ 4.4
10^7	10^4	1.6 M	316 k	5.1	4.7	~ 5.3
10^7	10^5	10.2 M	1.0 M	10.2	5.1	~ 5.6
10^7	10^7	29 M	10 M	2.9	6.1	~ 6.0

Read the ratio column: over five orders of magnitude in each axis it stays within $[0.2, 10]$ — a one-constant sqrt fit (say anchored at the central cells, $a \approx 2.6$) leaves residuals of only $0.1\times\text{--}4\times$, and the worst residuals sit at the extreme corners ($D = 1$ at huge N ; $D \approx N/b$, where the plateau has just vanished and the peak term crosses from $b \cdot D$ to N) where wall

time is dominated by fixed per-session costs (preamble, handshake, ~ 8 frames, thread/pipe synchronization in the bench harness) or by the $\Theta(D)$ body-handling term. Local log-log slopes of the exact sum, vs. sqrt’s uniform 0.5:

- in D at fixed $N = 10^6$: 0.75, 0.96, **0.47, 0.43**, 0.69 per decade ($D: 1 \rightarrow 10 \rightarrow 10^2 \rightarrow 10^3 \rightarrow 10^4 \rightarrow 10^5$) — *in the densest benchmark range the true slope is ≈ 0.45 , indistinguishable from 0.5*;
- in N at fixed $D = 10^3$: 0.80, **0.08**, 0.43 per decade ($N: 10^4 \rightarrow 10^5 \rightarrow 10^6 \rightarrow 10^7$) — the $10^5 \rightarrow 10^6$ decade is the smoking gun: the true cost is nearly flat there (it sits between staircase steps: $b^2 = 65,536$, $b^3 = 16.8\text{M}$), while sqrt demands $\times 3.16$.

Conclusion. $\sqrt{C \cdot D}$ and $\Theta(\min(bD, N)(1 + \log_b(N/D)))$ are empirically indistinguishable from wall time on diagonal or D -sweeps at grid scales; they separate cleanly on **fixed- D , N -sweeps across the $10^5 \rightarrow 10^6$ decade**, and on any *count*-based (rather than time-based) measurement.

8 Expected bandwidth in closed form

Theorem 2 gave the bandwidth’s shape. This section prices it: an explicit expectation $E[B](C, D, M)$ — total bytes on the wire, both directions, to reconcile two replicas sharing C live messages and differing in D leaves, with mean serialized message body M . (A `Message<T>` serializes byte-identically to its inner T , `message.rs:12-14`, so M is §3’s $\bar{s}$ as measured on the wire.) Split $D = D_{\text{live}} + D_{\text{red}}$ into leaves the counterparty must receive and leaves redacted on one side (§10.3); $N = C + D$ as always. Assumptions: A1 (uniform content addressing) and A8 (wire sizes); the $D = D_L + D_R$ split between the sides turns out not to matter, shown at the end.

providing: $33 + s_v + M$ bytes per live differing leaf. By A4 each differing leaf crosses the wire exactly once: a Left case ships in the same message, a Right case is requested and answered whole one message later (`partition.rs:91-142`), and disputed prefixes never ship content, only hashes. A leaf provided alone at pairing depth d costs, per the node encoding (`untyped.rs:406-443`; channel shapes at `message.rs:50-63`): its pairing `Prefix`, d raw bytes (the typed height pins the byte count, so there is no length prefix, `prefix.rs:127-156`); the node’s `prefix_len` byte plus its $32 - d$ compressed path bytes (the spine from depth d down to the leaf); its `Version`, s_v bytes on average; and its body, M bytes. The pairing prefix and the compressed spine *partition* the leaf’s 32 path bytes, so the path overhead is $d + 1 + (32 - d) = 33$ bytes — independent of where the leaf separated. When several differing leaves ship under one provided prefix, the shared spine bytes are paid once and branch bookkeeping appears instead (one `prefix_len` and one `count_minus_two` byte per node, one radix byte per child): the per-leaf path overhead ℓ then sits in $[2, 35]$, and sharing only helps; the singleton value $\ell = 33$ is the common case, since uniform paths rarely co-separate at $D \ll N$. Redacted leaves ship *nothing* here: both the Left arm and `answer_requested` filter through `Unknown::unknown` before anything is serialized (`partition.rs:91-142`, `232-239`; `unknown.rs:28-34`), so a redacted leaf is billed for descent — it counts in D , hence in W — and at most one `requested` record, but zero body bytes. In expectation:

$$E[\text{providing}] = D_{\text{live}} \cdot (\ell + s_v + M), \quad \ell \approx 33.$$

uncertain: W records of $h + \bar{k}$ bytes. §6 established that the records listed at depth k are exactly the occupied children of the disputed depth- $(k-1)$ parents: the wire carries each disputed parent's *full occupied sibling fan* — the Both arm explodes every grandchild, not only the differing ones (`partition.rs:243-254`) — and each depth is listed by exactly one side (the alternation; Theorem 2). The session-total record count, both directions summed, is therefore Theorem 1's W itself:

$$E[\text{\#records}] = W = \Theta(\min(b \cdot D, N) \cdot (1 + \log_b(N/D))).$$

A record at depth k is k raw prefix bytes plus the $h = 32$ -byte hash (A8); W 's mass sits within a level of the saturation depth j_N (the head and shoulder of §6's regime table), so the mean record size is $\bar{r} = h + \bar{k}$ with $\bar{k} \leq j_N + 1$: $\bar{r} \approx 34\text{--}38$ bytes for $N \leq 10^7$. The numbers below use $\bar{r} = 36$.

requested: lower-order. One bare prefix per separated they-only subtree: R records, $R \leq D$ (each names a subtree holding at least one differing leaf, and nothing is requested twice), each $\leq k^* + 1 \leq 33$ bytes, no hash. So $\leq 33 \cdot D$ bytes total — under the hash channel by a factor $\sim \bar{r} \cdot b/33 \approx 280$ in the dispute regime $D \leq N/b$, and never above the bodies' fixed per-leaf overhead. Lower-order everywhere.

Framing and handshake: $O(1)$. Two 25-byte raw preambles (`remote.rs:11-17`); two framed Handshakes carrying one Version each, $4 + s_v$ bytes framed (`message.rs:127-144`); at most 36 frames per session (§1) at 4 length bytes each (`remote.rs:26-33`); and a u32 length per channel vector, at most three per frame (`message.rs:4-5`). Total fixed cost

$$F \leq 2 \cdot 25 + 2 \cdot (4 + s_v) + 36 \cdot 4 + 36 \cdot 12 \approx 630 + 2 \cdot s_v \text{ bytes,}$$

independent of C , D , and M .

The expectation. Summing the four contributions:

$$E[B] = \underbrace{D_{\text{live}} \cdot (\ell + s_v + M)}_{\text{bodies}} + \underbrace{W \cdot (h + \bar{k})}_{\text{hashes}} + \underbrace{R \cdot \bar{k}_q}_{\text{requests}} + F,$$

with $\ell \approx 33$, $h = 32$, $\bar{k} \leq j_N + 1$, $R \leq D$, $\bar{k}_q \leq 33$, $F = O(s_v)$. Dropping the lower-order terms and absorbing the constants:

$$E[B] = \Theta\left(D_{\text{live}} \cdot (M + s_v) + \bar{r} \cdot \min(b \cdot D, N) \cdot (1 + \log_b(N/D))\right), \quad \bar{r} = h + \bar{k} \approx 36.$$

Two caveats, both already on the record. The hash term is the smooth envelope of §7's staircase, so realized values mid-step run several times below it. And when one side is empty (§10.2) the dispute machinery never engages: the hash term collapses to the Opening's $O(b)$ records and $E[B] \approx D \cdot (\ell + s_v + M) + F$ — the formula's hash envelope is then an overstatement, though the bodies dominate it there regardless.

Crossover: when do bodies overtake hashes? Set the two leading terms equal and solve for M :

$$M^* = \bar{r} \cdot \frac{W}{D_{\text{live}}} - (\ell + s_v).$$

On the smooth envelope with $D \leq N/b$ and $D_{\text{red}} = 0$, $W/D = b \cdot (1 + \log_b(N/D))$, so

$$M^* \approx 36 \cdot 256 \cdot (1 + \log_{256}(N/D)) \approx 17, 21, 25 \text{ KB at } N/D = 10^2, 10^3, 10^4$$

— the source of Theorem 2’s “~10–40 KB per differing message” figure (the envelope, $\pm$ Lemma 1’s occupancy constants). The realized figures, taking W from §7’s exact-sum table, agree with the envelope only where it is tight, $D \lesssim b$: there $M^* \approx 9$ –24 KB (the $D = 1$ rows and the $D = 10^2$ cells). For $D \gg b$ the head levels’ fans are shared among all D differing leaves and the shoulder fan is $N/b^k < b$ — the same staircase discount as §7 — and M^* falls to ~0.3–6 KB (e.g. $W/D = 22$ at $(N, D) = (10^6, 10^4)$, so $M^* \approx 0.7$ KB). Theorem 2’s headline survives with its regime named: **hash traffic dominates for bodies under ~10–25 KB when divergence is sparse ($D \lesssim b$), but only under ~0.3–6 KB when it is heavy ($D \gg b$).**

Numbers. W from §7’s table at $N = 10^6$ (so $C \approx N$), $D_{\text{red}} = 0$, $\bar{r} = 36$, $\ell = 33$, and a nominal $s_v = 16$ B; “share” is the hash channel’s fraction of $E[B]$ (requests and framing are negligible at this scale):

C/D	D	W	hash bytes	M^*	hash share of $E[B]$		
					$M = 256 \text{ B}$	$M = 4 \text{ KB}$	$M = 64 \text{ KB}$
$\sim 10^6$	1	527	19 KB	18.9 KB	98%	82%	22%
$\sim 10^4$	10^2	27 k	0.97 MB	9.7 KB	97%	70%	13%
$\sim 10^2$	10^4	220 k	7.9 MB	0.74 KB	72%	16%	1.2%
	9 10^5	1.08 M	39 MB	0.34 KB	56%	8.6%	0.6%

Read against the document’s running example: moving one differing message through a 10^6 -leaf tree costs ~19 KB of hashes (§4’s 18 KB) however small the body — and that constant is the radix-256 tradeoff of §10.6, not a fixable inefficiency.

Does the D_L/D_R split matter? For the totals, no: each live differing leaf ships exactly once whichever side holds it, and W already counts both directions’ listings. The only split-sensitive quantity in the derivation is Lemma 2’s pair count $D_L(C + D_R) + D_R(C + D_L) = DC + 2D_LD_R$, which ranges from DC (one-sided) to $DC + D^2/2$ (even split) — at most a factor 2 inside the $\log_b$, i.e. $\leq \log_{256} 2 \approx 0.13$ levels of separation depth. Per-direction loads do follow the split: the side holding D_L of the difference sends D_L of the bodies, while the alternation splits the hash records between the directions regardless.

9 An experiment that distinguishes them

All using existing infrastructure; no protocol changes.

1. **Byte counting (noise-free, decisive).** In `benches/gossip_grid.rs`, wrap the `PipeReader/PipeWriter` ends of `Wire` (`gossip_grid.rs:43–99`) in trivial

`std::io::Read/Write` adapters that increment `AtomicU64s`, and record bytes per session per cell. Bandwidth is $\Theta(36 \cdot W + \text{body terms})$ by Theorem 2 and is exactly measurable. Regress $\log(\text{bytes} - a \cdot D)$ on $\log D$ and $\log(N/D)$ over cells with $D \geq 100$: the sqrt model predicts elasticity ≈ 0.5 in $\log(N/D)$; Theorem 1 predicts ≈ 0 (a log-of-log). Frame *counts* from the same adapter (count 4-byte headers) verify Theorem 3 against $\frac{1}{2} \cdot \log_{256}(2DN) + c$ to ± 1 frame.

2. **The flat decade.** Fix *differing* = 1,000, *redacted* = 0, sweep *common* $\in \{10^3, 10^4, 10^5\}$ (cells already exist; extend `COMMON` in `benches/support/grid.rs:74` to 10^6 for the decisive decade, accepting the 10-sample floor). Predicted wall-time ratios *common* $10^5 \rightarrow 10^6$: sqrt $\times 3.16$; Theorem 1 $\times 1.2$ (the 0.08-slope decade). One afternoon of Criterion time.
3. **Direct hash-compare counts.** Drive two `local::Exchanges` in-process through the `#[cfg(test)] mirror::mirror` entry (`mirror.rs:337-353`) with grid-shaped trees and count `uncertain` entries (e.g. instrument in a test, or just count from the byte totals: $\text{entries} \approx (\text{uncertain bytes}) / (34 \dots 64)$). This measures W itself, bypassing the $\alpha \cdot D + \text{fixed}$ wall-time floor entirely.

10 Caveats and edge regimes

1. **Lazy memoization (first session after writes).** A batch of m local changes leaves cold `OnceLocks` on the $\Theta(m \cdot (1 + \log_b(N/m)))$ ancestor nodes of the changed paths (`untyped.rs:40-47`; memos survive Arc-sharing, so *only* changed paths are cold, `untyped.rs:100-123`). The first `hash()` then pays one branch preimage per cold node — up to $1 + 33 \cdot 256 \approx 8.4\text{KB}$ of BLAKE3 for a saturated branch, but only 34 bytes per compressed-spine level (`untyped.rs:283-287`; both conventions measured in `benches/hash_recon.rs`). This cost is additive, amortizes across sessions, and is excluded from the benches by `warm_caches()` (`gossip_grid.rs:154-160`).
2. **Bootstrap / one side empty ($D = N$).** No prefix has content on both sides, so there are *zero* disputes: the empty responder sends an empty `Opening` (`local.rs:337-341`), and the initiator's root-level drain Left-provides every root child in one message (`partition.rs:276-294`), filtered through `Unknown` against the empty version (everything survives). Cost: $\Theta(N)$ transfer and absorption, $O(1)$ rounds. Theorem 1's formula degrades gracefully to $\Theta(N)$ here, but the *mechanism* is different — the dispute machinery never engages.
3. **Deletion honoring.** A redacted leaf (present on A, deleted on B) keeps hashes differing along its path, so it costs descent exactly like a differing leaf — **count redactions in D** (the grid's throughput accounting already does: `grid.rs:115-127`). But it costs no body bytes: at separation, A's Left-provide is killed by the $O(1)$ ceiling check (`unknown.rs:28-34`) and A deletes its own copy (`partition.rs:232-239`) — deletion *propagates* through the version ceiling, with no tombstone and no transfer. A whole missing-but-dominated subtree dies the same way: one memoized comparison, no descent below it.
4. **Skew.** A1 is enforced by content addressing, not hoped for: because the version is folded into the leaf path (`path.rs:32-44`), even a million-message single-party version-batch

spreads uniformly — concentration under few prefixes would require engineered BLAKE3 outputs. (Contrast key-ordered Merkle trees, where a contiguous batch *does* cluster.) Even granting an adversary clustered paths, the depth ceiling caps the damage at $W \leq 32 \cdot b \cdot D$ and 18 RTT.

5. **Version sizes.** ITC Versions are not constant-size: they ride the handshake, every provided leaf, and every `ceiling/floor` $\leq$ comparison (`unknown.rs:31, 46-49`), contributing the s_v factors in Theorems 1–2. For long-lived universes with many party splits, s_v is the term to watch, orthogonal to the tree combinatorics.
6. **The radix constant is the design tradeoff.** Per disputed node the protocol ships a full sibling fan (up to 256×36 ish bytes ≈ 9 KB) to buy a 2.4-decade depth reduction per level. $b = 256$ puts every realistic deployment within ~ 3 levels of separation (RTT-optimal) at the cost of the $\times b$ in W — which is exactly the constant that made a $\sqrt{C \cdot D}$ fit land in the right numeric range in the first place.